datacenter space but immediately your Chief Financial Officer (CFO) would remind you that you are a startup and you don't have millions to spend on just IT infrastructure; you need funds to grow many other business units like marketing, sales, workplace resources and you have to hire more engineers too. You could argue that you have saved a lot of money on software by using open source and instead of buying expensive licenses for commercial technologies you used Linux, MySQL, JBoss and Python/PHP to develop your applications. But all the hardware required to run your stuff is still a big expense, and not to forget, you will need a team of tech savvy team of system administrators who can effectively manage this infrastructure and keep them updated, backed up, patched and secure.

On the other hand, business applications that your company needs to run for its own operations are also not absolutely straightforward. Being a technology savvy company you may decide to develop your own systems for managing HR, Finance, Email etc., but that may not be the best use of your engineering resources. However right from evaluating a business application to successfully deploying it may be very time consuming and expensive.

Now here is where cloud computing can play a big role in giving you a headstart and provide you a platform that could eliminate or at least minimise many of these technical and business challenges.

Cloud computing in essence is a utility model of consuming hardware and software resources, much like how you consume electricity at your home. It can provide you a platform where you practically have limitless supply of hardware and software, available to you as a service where at a push of a button you can get everything you need to develop your application, deploy it for anyone in the world to use it over the internet, and manage the entire lifecycle of the application while being billed for only how much you use it.

You may ask, so what is the big deal? Imagine if you had to build a nuclear power power plant and had to have the technical knowhow and resources to operate one in order to have electricity supply at your home to power your TV, XBOX and PC. Believe it or not, but most of the large scale enterprise computing happens this way, even today. An automobile manufacturing company may employ more software engineers and system administrators, have bigger datacenter and spend more on buying software and hardware than an equally large software development company!

Introducing Cloud

There are three broad categories under which cloud services could be categorized, let's understand each one:

Software as a Service (SaaS)

Probably, the most popular method of using cloud services, Software as a Service lets you use a software directly on a subscription basis, eliminating various technical challenges on the way. Many business applications today are available for use through a Software as a Service model. At BestEverything.in, you will need to provide all your employees with a @BestEverything.in email. You may need to provide them with productivity software for word and spreadsheet processing. Everyone in the HR department will need a HR management system, the sales and marketing team may need a customer relationship management software. Your web administrator may need some sophisticated software to track and analyse the trend of visitors at BestEverything.in, etc. You could very simply implement the solutions for these needs through plethora of SaaS offerings available over the internet. For example, it could take many weeks to implement a mailserver on your own, but companies like Google, Microsoft and Zoho can provide you a SaaS based mail and messaging service through which you could have a fully functional email service rolled out for the entire company in less than an hour. Many As SaaS model takes away the technical complexity of installation, configuration and ongoing maintenance of the software, BestEverything can implement best-in-class business applications (think SAP) from day one of its operations and follow best-in-class business processes that will make running BestEverything as efficient as possible. And it will automatically update to its latest version.

Apart from saving you time and technical efforts, SaaS also saves money. Software delivered through SaaS is charged per user and therefore only people in finance who use the finance software will add to the cost. However, software delivered through SaaS may have limited customizability and sometime you may be forced to go to a new version of the software even if you do not want to do so. But for a startup, it is a great way to get started and scale the business without wasting time, effort and money in implementing different business applications.

Infrastructure as a Service (IaaS)

In order to develop and test the BestEverything.in, the development team has raised a request for 10 virtual machines and 5TB of storage capacity. For running the production environment, the team needs 2 database servers, 4 application servers, and 6 webservers. Minimum of two load balancers are required to spread the incoming traffic across the web servers. Moreover, the exact setup should be provisioned in a second datacenter somewhere nearby the first one

Some Open Source business applications to consider for your Startup

Enterprise Resource Planning	Odoo	https://www.odoo.com/
Human Resource Management	Orange HRMS Live	http://www.orangehrm.com/OrangeHRMLive.php
Customer Relationship Management	SugerCRM	http://www.sugarcrm.com/
Enterprise Portal and Collaboration	Liferay	http://www.liferay.com/

<Exploring cloud is relatively easy, as many popular cloud service providers have options to help you try their IaaS and PaaS solution before you make any purchase or investments. For example, Amazon offers its users a free usage tier, through with a new user can spin small VMs on their Elastic Cloud (EC2) for no charge for up to a total of 750 hours. Similarly, Microsoft Azure provides a one month free trial of their cloud options.>